The real world (the curse of reality): irregular objects · long-tail scenarios · closed-loop shift · train–deploy gaps

observations ↓ ↑ *actions*

Stage 1 · Perceive — RQ1

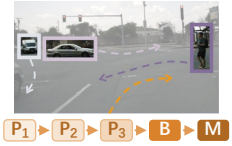
Chapter 2
OccNet, OpenOcc, OpenScene



Voxels capture what rigid boxes miss; OpenOcc and OpenScene benchmark occupancy at scale

Stage 2 · Reason — RQ2

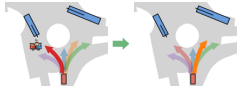
Chapter 3
DriveLM-Agent, DriveLM-Data



Graph-structured VQA grounds a VLM from key objects to behaviour and motion

Stage 3 · Act — RQ3

Chapter 4
Centaur, test-time training



A test-time training step on Cluster Entropy removes the colliding plan (NAVSIM)

Chapter 5
χ_0, resource-aware manipulation



Train–deploy alignment for bimanual garment manipulation on real robots

Unified Perspective
Chapter 6 (RQ4)

Five elements of a self-correcting physical agent: (1) an open, queryable 3D representation · (2) fast and slow decisions (ETA*, FLARE*) · (3) a competence monitor · (4) adaptation at deployment · (5) a learning loop (PerturBot*, self-directed learning*)